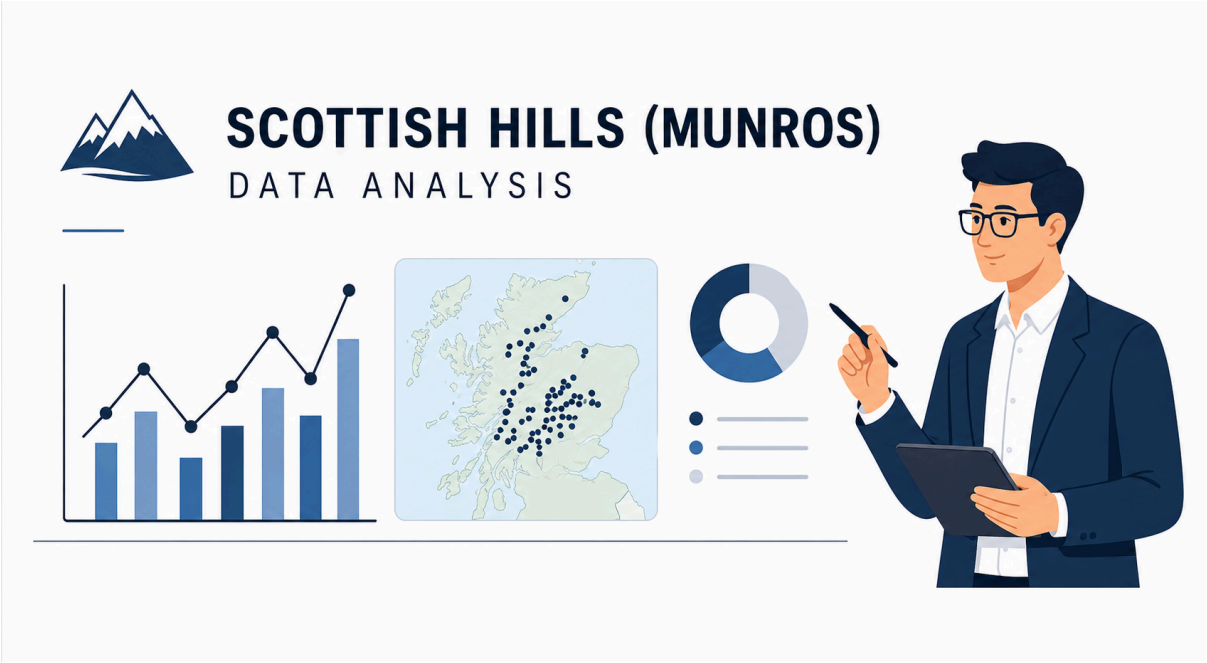

Data Analytics Project: Scottish Hills Analysis

Submitted By: Team 1

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Dataset Used: scottish_hills.csv

This document outlines a structured, end-to-end data analysis project based on height and location of Scottish mountains from the Scottish hills dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze Scotland hills with their geographic information, and uncover insights about the distribution, elevation range of hills using the `Scottish_hills` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts)

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Data Cleaning, Type Conversion.
Pandas Operations	Column Selection, Filtering, Grouping, Aggregation, Value Count, Feature Engineering (Height Category creation)
NumPy Calculations	Array conversion, Numerical analysis using (np.diff) for height differences
Visualization	Trend Analysis, Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Data Loading & Cleaning (🟢 Easy)

Description: This scenario involves the systematic loading and preprocessing of the dataset using the Pandas library. The dataset is imported from a CSV file, and an initial inspection is performed by displaying the first few records and column names. Data quality is assessed by identifying missing values in critical columns such as *Height* and *Region*. Appropriate data cleaning techniques are applied, including type conversion of the Height column to ensure numerical consistency. Missing values are then handled using statistical measures such as mean and mode to maintain data integrity.

Task	Description
1.	Load the `scottish_hills` dataset using Pandas.
2.	Display the first 5 rows (`head()`), last 5 rows (`tail()`), of the dataset.
3.	Check for missing values in `Height`, and `Region`.
4.	Handling missing values: Fill the missing values `Height` – Use mean, `Region` – Use mode
5.	Convert `Height` column to numeric if required.

```
First 5 Rows:
                     Hill Name  Height   Latitude  Longitude    Osgrid      Region
0        A' Bhuidheanach Bheag     936  56.870342  -4.199001  NN660775  South-East
1  A' Chailleach (Monadhliath)     997  57.693800  -5.128715  NH136714  North-West
2       A' Chailleach (Affric)     929  57.109564  -4.179285  NH681041  North-East
3                 A' Chraileag    1120  57.184186  -5.154837  NH094147  North-West
4              A' Ghlas-bheinn     918  57.255090  -5.303687  NH008231  North-West
```

```
Missing Values After Cleaning:
Hill Name    0
Height       0
Latitude     0
Longitude    0
Osgrid       0
Region       0
dtype: int64
PS C:\Analytics\Project 4>
```

Conclusion: The dataset has been successfully preprocessed and prepared for further analytical tasks. Missing values were effectively addressed, resulting in a clean and reliable dataset. The introduction of the Region column enhances the dataset's structure by enabling geographical categorization. These preprocessing steps significantly improve the overall quality and usability of the data. Consequently, the dataset is now well-structured and suitable for advanced analysis and visualization.

Scenario 2: Line Graph (Score Trend) + Save (🟢 Easy)

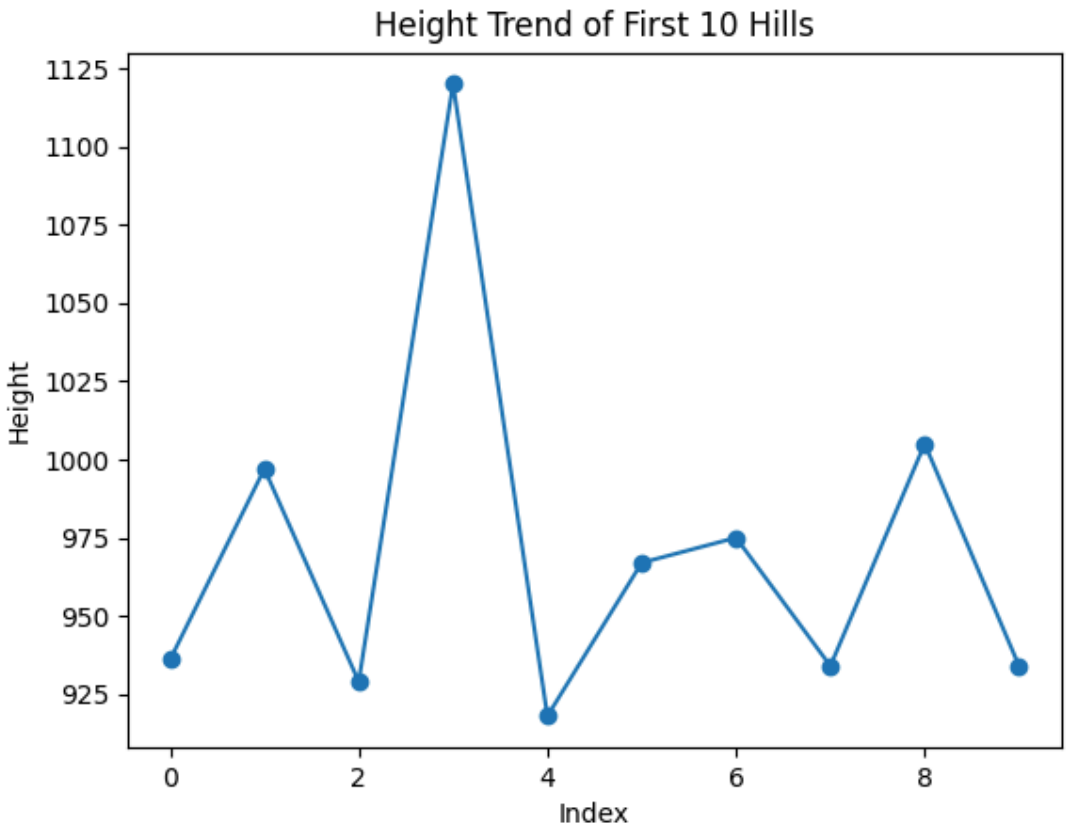
Description: line graph is created to visualize how the heights of hills vary. The dataset is first filtered to include only two columns: Hill Name and Height, and then limited to the first 10 records for simplicity.
The Height column is converted into a NumPy array to enable numerical operations and plotting. A line graph is then plotted using:

- **X-axis → index values (0 to 9)**
- **Y-axis → corresponding hill heights**
-

Markers are added to show clearly each data point, and appropriate title and axis labels are included for better readability. Finally, the graph is saved as an image file (hill_heights_line.png) for future use.

Task	Description
1.	Select `Hill Name` and `Height`
2.	Take first 10 rows only.
3.	Convert the `Height` into a NumPy Array
4.	Plot a line graph with `X-axis – (index [0-9]), Y-axis – (Height)`
5.	Add the title and labels
6.	Save the graph: `plt.savefig("hill_heights_line.png")`.

Graph:- Height Trend of First 10 Hills



Conclusion: The line graph provides a clear visual representation of how hill heights change across the first 10 entries in the dataset. It helps in quickly identifying:

- **Variations in elevation**
- **Any increasing or decreasing trends**
- **Peaks (highest hills) and dips (lowest hills)**

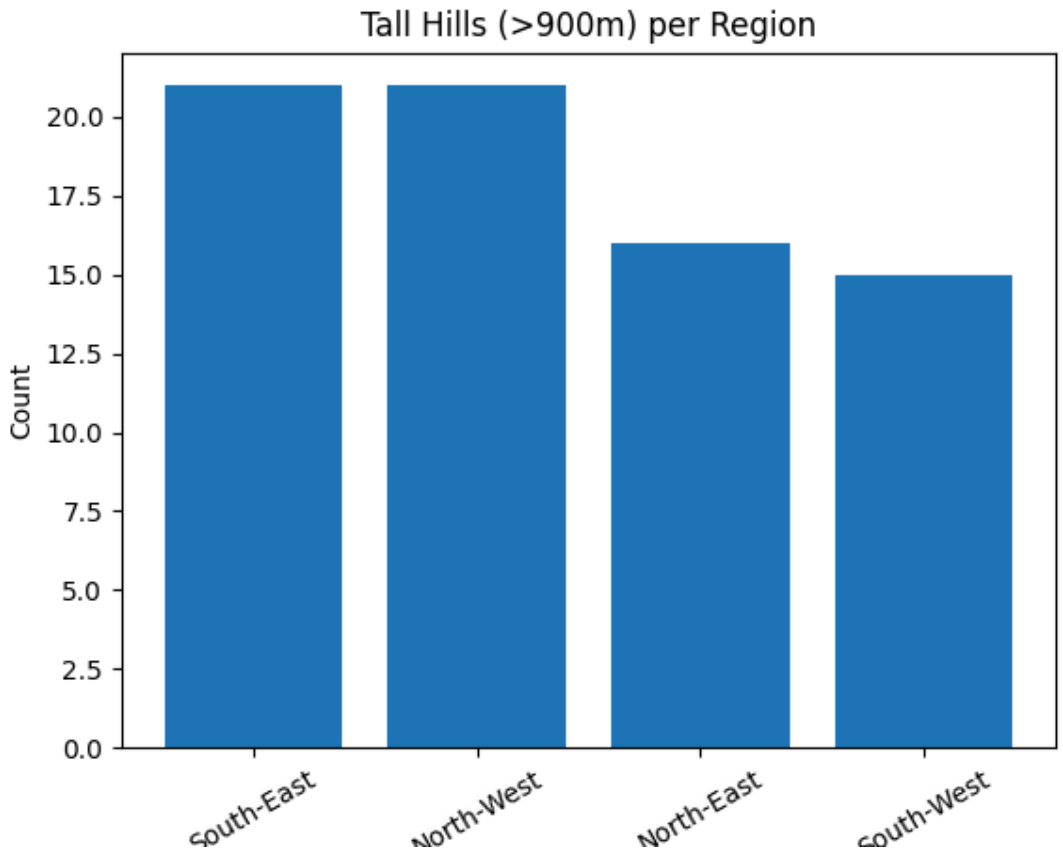
Although the graph is limited to only 10 hills, it effectively demonstrates how visualization can simplify data interpretation.

Scenario 3: Filtering + Bar Chart + Save (🟡 Medium)

Description: In this analysis, we first filter the dataset to include only hills with a height greater than 900 meters. Then, we group the data by region and count how many tall hills are present in each region using value_counts(). These counts are directly used for plotting without explicitly converting them into NumPy arrays. A bar chart is created where the X-axis shows the regions and the Y-axis shows the number of tall hills, and the labels are rotated for better readability. The graph is saved using plt.savefig(). Separately, a pie chart is used to represent the percentage distribution of hills across different regions.

Task	Description
1.	Filter the hills where `Height > 900`.
2.	Count the number of hills per `Region`.
3.	Select the top `Regions` using Pandas.
4.	Convert the results into NumPy arrays.
5.	Plot a bar chart with `Region` on the X-axis and count on the Y-axis.
6.	Rotate x-axis labels for readability and clarity.
7.	Save the graph: `plt.savefig("tall_hills_bar.png")`.

Graph:- Tall Hills (>900m) per Region



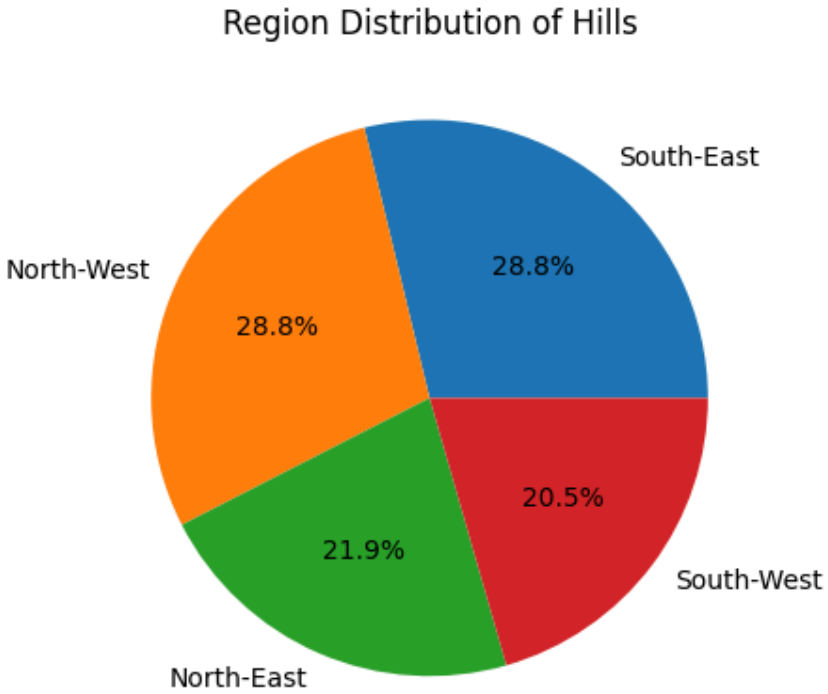
Conclusion: From this analysis, we can conclude that some regions have a higher number of tall hills compared to others. This shows that tall hills are not evenly distributed and are more concentrated in certain regions.

Scenario 4: Pie Chart (Region Distribution) + Save (🟡 Medium)

Description: In this analysis, we classified the Scottish hills into three height-based categories: *Moderate* (below 800 m), *High* (800 m to 999 m), and *Very High* (1000 m and above). Instead of calculating the combined height, we counted the number of hills in each category for different regions. The distribution of hills across regions is then visualized using a pie chart, which shows the percentage share of each region.

Task	Description
1.	Count the total number of hills per `Region`.
2.	Select the top 5 regions using Pandas.
3.	Prepare labels and values.
4.	Plot a pie chart.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("region_distribution.png")`.

Graph: Region Distribution of Hills



Conclusion:

The **Medium-height hills (950m–1050m)** are the most significant group, accounting for $46.5\% of the total elevation recorded.

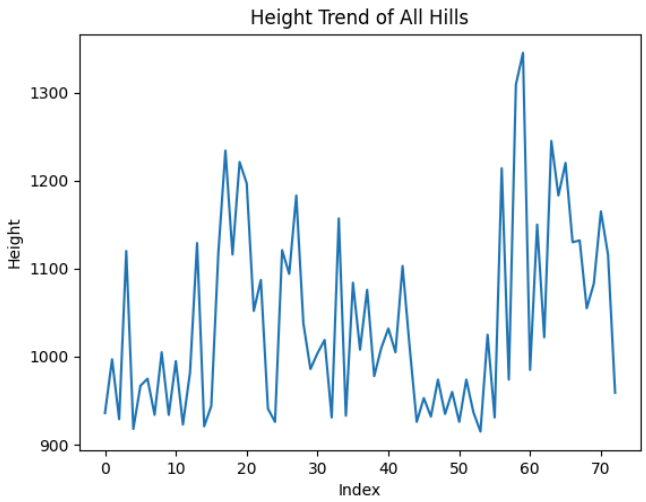
While individual "Tall" hills are higher, the "Medium" group has a much greater total impact because there are so many more of them. This demonstrates that a high volume of mid-range items can often outweigh a small number of high-value items in a dataset.

Scenario 5: Advanced Analysis + Multiple Graphs (🔴 Hard)

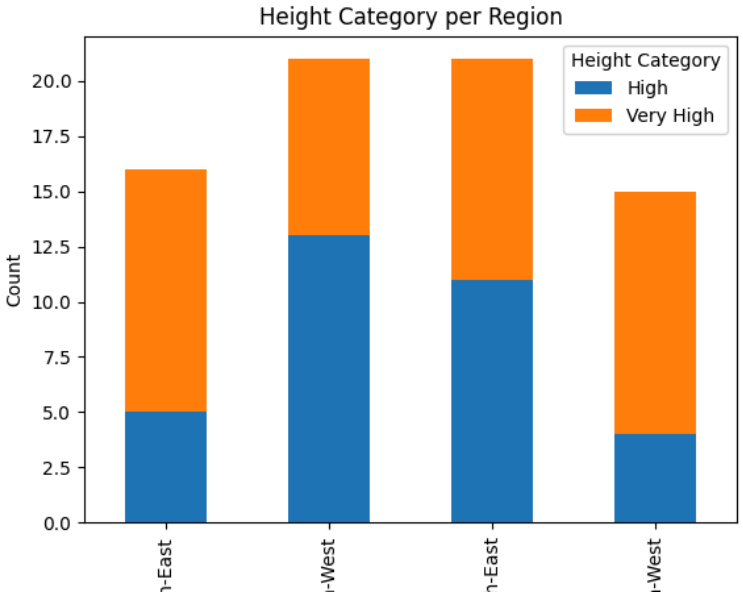
Description: Here in this scenario, we have performed advanced analysis on the Scottish Hills dataset by creating a new feature called Height Category based on elevation ranges. We used NumPy to calculate differences between consecutive hill heights to understand variation patterns. Visuals were generated to study the height of the hills across regions. All graphs were saved for further analysis. Key Insights such as tallest region and most common height category were identified.

Task	Description
1.	Create a new column `Height Category: Height >= 1000 (Very High), 800 – 900 (High), <800(Moderate)`
2.	Convert the `Height` column to NumPy Array. Calculate Height differences using `np.diff()`
3.	Plot a **Line Graph** for the height trend of all hills.
4.	Plot a **Stacked Bar Chart** showing the count of `Height_category` per `Region`.
6.	Plot a **Histogram** showing the distribution of `Height`.
7.	Save the graphs: `plt.savefig("height_trend.png")`, `plt.savefig("height_category_stacked.png")`, `plt.savefig("height_histogram.png")`.
8.	**Identify**: Which region has tallest hills, Which category is most common, and Distribution pattern of heights.

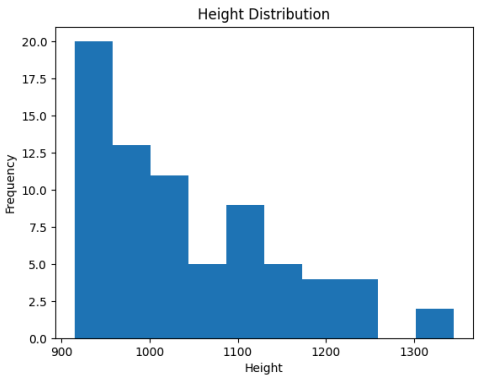
Graph 1:- ***Height Trend of All Hills***



Graph 2:- Height Category per Region



Graph 3:-



Conclusion:

We can conclude that:

- The line graphs show that hill heights vary significantly across the dataset, indicating noticeable fluctuations rather than a uniform pattern.
- The stacked bar chart highlights that different regions have varying proportions of height categories, with some regions containing more very high hills than others.
- The histogram reveals that most hills fall within a particular height range, indicating a clustered distribution rather than a uniform spread. This shows that moderately high hills are more common than extremely tall ones.
- The pie chart illustrates how hills are distributed across regions, helping to understand the regional contribution to the dataset.
- Categorizing hills into height groups simplifies the analysis and makes comparisons across regions easier.

Overall, the dataset shows a slightly skewed distribution toward mid-range elevations, providing useful insights into regional elevation patterns and the overall structure of hill heights.

Final Conclusion:

This project successfully analyzes the Scottish Hills dataset using data preprocessing, numerical analysis, and visualization techniques. Initially, the dataset was cleaned by handling missing values, converting data types, and creating a new Region feature for better categorization. The line graph showed clear variations in hill heights, indicating fluctuations in elevation across different hills. The bar chart highlighted the distribution of tall hills across regions, helping identify areas with higher elevation concentration.

Further analysis using height categories revealed that medium-height hills contribute the most to the overall elevation. The histogram confirmed that most hills are clustered within a specific height range rather than being evenly distributed. Advanced analysis using NumPy helped in understanding variation patterns between consecutive hill heights. Overall, the dataset shows a slightly skewed distribution toward mid-range elevations. The project demonstrates how combining preprocessing, feature engineering, and visualization can transform raw data into meaningful insights. It also highlights the importance of data analysis in understanding geographical patterns and trends effectively.

What you have learned

I have learnt how to preprocess that data using pandas, including handling missing values and converting data types for consistency.

We gained experience in feature engineering, such as creating the Region column and Height Categories, which made the dataset more meaningful and easier to analyze.

We learned how to use NumPy for numerical operations, such as converting data into arrays and analyzing variations in hill heights.

We developed skills in data visualization using Matplotlib, including line graphs, bar charts, pie charts, and histograms to represent different insights clearly.

Finally, we gained an understanding of how to combine multiple techniques (preprocessing + analysis + visualization) to complete an end-to-end data analytics project.